# SHARP COEFFICIENTWISE POSITIVITY FOR A MATRIX TURÁN DETERMINANT OF ${}_0F_1$

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ABSTRACT. We prove a sharp coefficientwise positivity theorem for a matrix Turán determinant of the normalized hypergeometric function ${}_0F_1$. From the reciprocal-gamma normalization $Z(a,\lambda) = {}_0F_1(;a;\lambda)/\Gamma(a) = \sum_{k\geq 0} \lambda^k/(k!\,\Gamma(a+k))$, $a > 0$, we form symmetric $2 \times 2$ matrices $\mathcal{T}_\kappa(a,\lambda)$ from parameter and Euler derivatives of $Z$, and prove that every positive-degree Maclaurin coefficient of $\Delta^{(\kappa)} = \det \mathcal{T}_\kappa$ is strictly positive for all $a > 0$ if and only if $\kappa \geq 1$. At the endpoint $\kappa = 1$, $\Delta^{(1)}(a,\cdot)$ is strictly absolutely monotone even though the first non-constant coefficient matrix can be indefinite; its potentially negative degree-two determinant contribution is offset by a manifestly positive Hilbert-space decomposition. Via the identity $I_{a-1}(2\sqrt{\lambda}) = \lambda^{(a-1)/2} Z(a,\lambda)$, an explicit diagonal congruence yields a positive-definite Bessel–Schur matrix and a mixed order–argument inequality for $I_\nu$, for every $\nu > -1$; we also fix the exact uniform deformation threshold and, for each fixed $\nu$, the least argument-independent correction. Beyond $\nu > -1$ we exhibit explicit regimes where coefficientwise or pointwise positivity fails.

## 1. INTRODUCTION

Turán inequalities for modified Bessel functions are closely connected with order monotonicity, logarithmic derivatives, and total positivity; see [1, 2, 4, 6, 19]. A complementary literature proves scalar log-concavity and coefficientwise Turán inequalities for reciprocal-gamma series [8, 9, 10]. Here we study the normalized series

$$Z(a,\lambda) = \sum_{k\geq 0} \frac{\lambda^k}{k!\Gamma(a+k)}$$

through a coupled $2 \times 2$ matrix involving parameter and Euler derivatives.

A single theorem underlies the paper: the sharp coefficientwise classification of Section 2. The Bessel inequality is its corollary, while the optimality analysis determines the exact $\kappa$-threshold and the least $z$-independent correction on the positive-series domain $a > 0$ (equivalently $\nu > -1$), and the negative-order analysis exhibits explicit regimes where coefficientwise positivity of $\Delta$ or pointwise positivity of the Bessel–Schur determinant fails. The determinant is a normalized differential Schur complement, distinct from scalar parameter-shift Turánians and from total-positivity evaluation minors for $I_s(x)$ [5, 15] (compared in Section 9). The contributions are as follows.

(i) We embed the matrix in a one-parameter family and determine the exact uniform threshold: coefficientwise positivity for every $a > 0$ holds precisely for $\kappa \geq 1$ (Sections 2 and 7).

(ii) From an asymmetric Chu–Vandermonde convolution we derive exact coefficient formulas and interpret them through a finite conditional reciprocal-gamma law (Sections 4 and 5).

(iii) At the sharp endpoint we construct explicit $\ell^2$ Gram representations for every stable coefficient matrix. The only coefficient matrix that can be indefinite is $M_1$ when $0 < a < a_*$, where $a_*$ is the unique positive solution of $(4a-1)\psi_1(a) = 4$ ($a_* \approx 0.369$); its

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only potentially negative determinant contribution is the self-pairing in $\Delta_2$, which has a separate manifestly positive Hilbert-space factorization (Sections 6 and 7).

(iv) The endpoint theorem is transported by an explicit diagonal congruence to a positive-definite Bessel–Schur matrix and a mixed order–argument inequality for every $\nu > -1$. We determine both the sharp deformation threshold and, for each fixed order $\nu > -1$, the least $z$-independent correction (Sections 3 and 8).

(v) Beyond the positive-series domain, the leading small-argument term remains positive away from the poles, but coefficientwise positivity of $\Delta$ fails on $-3 < \nu < -2$ and the pointwise Bessel inequality fails near every negative integer $-n$, $n \geq 2$ (Section 8).

## 2. Main results

Throughout, $a > 0$, $\lambda \geq 0$, and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. We use

$$Z(a,\lambda) = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!\Gamma(a+k)} = \frac{{}_0F_1(;a;\lambda)}{\Gamma(a)}. \tag{2.1}$$

The series in (2.1) and all fixed-order $a$- and Euler derivatives used below converge locally uniformly on $\{\Re a > 0\} \times \mathbb{C}$: on a compact set $\partial_a^j \left[1/\Gamma(a+k)\right]$ is $1/\Gamma(a+k)$ times a polynomial in the polygamma values $\psi^{(0)}, \ldots, \psi^{(j-1)}(a+k)$, of at most polynomial–logarithmic growth in $k$, while $\lambda\partial_\lambda$ contributes powers of $k$; each differentiated series is therefore dominated by a constant multiple of $\sum_k |\lambda|^k/k!$, so termwise differentiation is valid. Put $\Theta_\lambda = \lambda\partial_\lambda$ and $g = \psi_1(a)$, where $\psi = \psi^{(0)}$ and $\psi_1 = \psi^{(1)}$ are the digamma and trigamma functions. In the Bessel variable we use the distinct notation $\Theta_z = z\partial_z$. Subscripts denote application of the indicated operator, so $Z_a = \partial_a Z$, $Z_{\Theta_\lambda} = \Theta_\lambda Z$, and $Z_{a\Theta_\lambda} = \partial_a \Theta_\lambda Z$. Define

$$A = Z_a^2 - ZZ_{aa}, \tag{2.2}$$

$$B = Z^2 + ZZ_{a\Theta_\lambda} - Z_a Z_{\Theta_\lambda}, \tag{2.3}$$

$$C_\kappa = Z^2 + g\big(\kappa Z Z_{\Theta_\lambda} - ZZ_{\Theta_\lambda\Theta_\lambda} + Z_{\Theta_\lambda}^2\big), \tag{2.4}$$

$$\mathcal{T}_\kappa = \begin{pmatrix} A & \sqrt{g}\,B \\ \sqrt{g}\,B & C_\kappa \end{pmatrix}, \qquad \Delta^{(\kappa)} = \det \mathcal{T}_\kappa = AC_\kappa - gB^2. \tag{2.5}$$

At $\kappa = 1$ write $C = C_1$, $\mathcal{T} = \mathcal{T}_1$, and $\Delta = \Delta^{(1)}$. A function on $(0, \infty)$ is *strictly absolutely monotone* if it and all of its derivatives are strictly positive there. Here *uniform* means that a single threshold in $\kappa$ works for every $a > 0$; it does not assert a positive lower bound uniform in $a$.

**Theorem 2.1** (Sharp coefficientwise classification)**.** *Uniform coefficientwise positivity holds exactly for $\kappa \geq 1$: for every $a > 0$ and $\kappa \geq 1$,*

$$\Delta^{(\kappa)}(a,\lambda) = \sum_{n\geq 1} \Delta_n^{(\kappa)}(a)\lambda^n, \qquad \Delta_n^{(\kappa)}(a) > 0, \tag{2.6}$$

*whereas for $\kappa < 1$ the linear coefficient is negative for all sufficiently small $a > 0$. Consequently, $\Delta(a, \cdot)$ is strictly absolutely monotone on $(0, \infty)$ and $\mathcal{T}(a, \lambda) \succ 0$ for $a, \lambda > 0$.*

The series in (2.6) starts at degree one because

$$A(a,0) = \frac{g}{\Gamma(a)^2}, \qquad B(a,0) = C_\kappa(a,0) = \frac{1}{\Gamma(a)^2},$$

so $\mathcal{T}_\kappa(a,0) = \Gamma(a)^{-2} \left(\begin{smallmatrix} g & \sqrt{g} \\ \sqrt{g} & 1 \end{smallmatrix}\right)$ has rank one and $\Delta^{(\kappa)}(a,0) = 0$.

**Corollary 2.2** (Explicit quantitative lower bound)**.** *For every $a, \lambda > 0$,*

$$\Delta(a,\lambda) > \Delta_1(a)\lambda = \frac{2\big(a\psi_1(a) - 1\big)}{a^3\Gamma(a)^4}\,\lambda > \frac{\lambda}{a^4\Gamma(a)^4}. \tag{2.7}$$

*Proof.* The exact value of $\Delta_1$ follows from (7.5) in Section 7. The first strict inequality follows from Theorem 2.1. Finally, (6.2) implies $a\psi_1(a) - 1 > (2a)^{-1}$. $\square$

For $\nu > -1$ and $z > 0$, the identity (3.1) and the positive coefficients of $Z(a,\lambda)$ show that $I_\nu(z) > 0$. Thus the following real logarithmic derivatives are well defined:

$$G_\nu = -\partial_\nu^2 \log I_\nu, \tag{2.8}$$

$$P_\nu = \partial_\nu(\Theta_z \log I_\nu), \tag{2.9}$$

$$H_\nu = 2(\Theta_z \log I_\nu - \nu) - \Theta_z^2 \log I_\nu, \tag{2.10}$$

and

$$D_\nu = G_\nu \left( H_\nu + \frac{4}{\psi_1(\nu+1)} \right) - (1+P_\nu)^2. \tag{2.11}$$

The shifts in $1 + P_\nu$ and $4/\psi_1(\nu+1)$ are exactly those produced by the diagonal congruence (3.5). For $\kappa \in \mathbb{R}$, define the deformation

$$H_\nu^{(\kappa)} = 2\kappa(\Theta_z \log I_\nu - \nu) - \Theta_z^2 \log I_\nu, \tag{2.12}$$

$$D_\nu^{(\kappa)} = G_\nu \left( H_\nu^{(\kappa)} + \frac{4}{\psi_1(\nu+1)} \right) - (1+P_\nu)^2. \tag{2.13}$$

At $\kappa = 1$ these reduce to $H_\nu$ and $D_\nu$. We call this the *fixed-correction deformation* because the $z$-independent correction $4/\psi_1(\nu+1)$ is held fixed while the argument-curvature term is deformed.

**Theorem 2.3** (Mixed Bessel–Schur inequality)**.** *For $\nu > -1$ and $z > 0$, $D_\nu(z) > 0$, equivalently*

$$G_\nu(z) \left( H_\nu(z) + \frac{4}{\psi_1(\nu+1)} \right) > (1+P_\nu(z))^2. \tag{2.14}$$

**Corollary 2.4** (Positive Bessel–Schur matrix)**.** *For $\nu > -1$ and $z > 0$,*

$$\begin{pmatrix} G_\nu(z) & 1+P_\nu(z) \\ 1+P_\nu(z) & H_\nu(z) + 4/\psi_1(\nu+1) \end{pmatrix} \succ 0.$$

*As $z \downarrow 0$, the matrix-valued function admits a continuous extension whose value is the rank-one matrix $\begin{pmatrix} g & 2 \\ 2 & 4/g \end{pmatrix}$, where $g = \psi_1(\nu+1)$.*

**Proposition 2.5** (Sharpness of the Bessel correction and deformation)**.** *For each fixed $\nu > -1$, $4/\psi_1(\nu+1)$ is the least real constant $R$ such that*

$$G_\nu(z)\big(H_\nu(z) + R\big) > (1+P_\nu(z))^2 \qquad \text{for every } z > 0.$$

*Moreover, for the fixed-correction deformation $D_\nu^{(\kappa)}$ defined in* (2.13),

$$D_\nu^{(\kappa)}(z) > 0 \qquad \text{for every } \nu > -1 \text{ and } z > 0$$

*holds if and only if $\kappa \geq 1$.*

Thus the correction is sharp separately for each fixed $\nu$, whereas the deformation threshold is sharp uniformly over the full domain $\{(\nu, z) : \nu > -1,\ z > 0\}$.

## 3. Exact reduction from ${}_0F_1$ to the Bessel inequality

The standard identity

$$I_{a-1}(2\sqrt{\lambda}) = \lambda^{(a-1)/2} Z(a,\lambda) \tag{3.1}$$

with $L = \log Z$, $z = 2\sqrt{\lambda}$, and $\nu = a - 1$ gives

$$\Theta_z \log I_\nu = \nu + 2L_{\Theta_\lambda}, \qquad P_\nu = 1 + 2L_{a\Theta_\lambda}, \tag{3.2}$$

$$G_\nu = -L_{aa}, \qquad H_\nu = 4(L_{\Theta_\lambda} - L_{\Theta_\lambda\Theta_\lambda}). \tag{3.3}$$

Moreover,

$$\frac{A}{Z^2} = -L_{aa}, \qquad \frac{B}{Z^2} = 1 + L_{a\Theta_\lambda}, \qquad \frac{C}{gZ^2} = \frac{1}{g} + L_{\Theta_\lambda} - L_{\Theta_\lambda\Theta_\lambda}. \tag{3.4}$$

The Bessel–Schur matrix is therefore the congruence

$$\begin{pmatrix} G_\nu & 1+P_\nu \\ 1+P_\nu & H_\nu + 4/g \end{pmatrix} = \frac{1}{Z^2}\begin{pmatrix} 1 & 0 \\ 0 & 2/\sqrt{g} \end{pmatrix} \mathcal{T}(a,\lambda) \begin{pmatrix} 1 & 0 \\ 0 & 2/\sqrt{g} \end{pmatrix}, \tag{3.5}$$

whose determinant is

$$D_{a-1}(2\sqrt{\lambda}) = \frac{4\Delta(a,\lambda)}{gZ(a,\lambda)^4}. \tag{3.6}$$

This reduces Theorem 2.3 and Corollary 2.4 to the endpoint $\kappa = 1$ of Theorem 2.1; the proof is completed in Section 8. The following calculations supply the two sharpness ingredients. Their formal use is deferred to Section 8, after positivity of $D_\nu$ has been established.

Replacing $4/\psi_1(\nu+1)$ in (2.14) by a $z$-independent $R(\nu)$ and writing $D_{\nu,R}$ for the resulting determinant,

$$D_{\nu,R}(z) \longrightarrow \psi_1(\nu+1)R(\nu) - 4 \qquad (z \downarrow 0),$$

so validity for every $z > 0$ forces $R(\nu) \geq 4/\psi_1(\nu+1)$. Conversely $D_{\nu,R} = D_\nu + G_\nu\big(R(\nu) - 4/\psi_1(\nu+1)\big) > 0$ whenever $R(\nu) \geq 4/\psi_1(\nu+1)$, since $D_\nu > 0$ and $G_\nu = A/Z^2 > 0$ for $\nu > -1$, $z > 0$. Hence $4/\psi_1(\nu+1)$ is the exact least $z$-independent correction.

For the fixed-correction deformation (2.12)–(2.13), the same calculation gives

$$D^{(\kappa)}_{a-1}(2\sqrt{\lambda}) = \frac{4\Delta^{(\kappa)}(a,\lambda)}{gZ(a,\lambda)^4}. \tag{3.7}$$

A direct first-order expansion,

$$Z(a,\lambda) = \frac{1}{\Gamma(a)}\left(1 + \frac{\lambda}{a} + O(\lambda^2)\right),$$

together with the corresponding termwise parameter and Euler derivatives in (2.2)–(2.4), gives

$$\Delta^{(\kappa)}(a,\lambda) = \frac{2}{a^3\Gamma(a)^4}\left[\frac{\kappa-1}{2}(ag)^2 + ag - 1\right]\lambda + O(\lambda^2). \tag{3.8}$$

Using (3.7) and $\lambda = z^2/4$ therefore yields

$$D^{(\kappa)}_{a-1}(z) = \frac{2}{a^3 g}\left[\frac{\kappa-1}{2}(ag)^2 + ag - 1\right] z^2 + O(z^4). \tag{3.9}$$

For $\kappa < 1$, the bracket is negative for all sufficiently small $a > 0$, because $ag \to +\infty$ as $a \downarrow 0$; thus (3.9) proves necessity of $\kappa \geq 1$. Sufficiency follows from Theorem 2.1 through (3.7). Hence the fixed-correction family is uniformly valid exactly for $\kappa \geq 1$.

## 4. Reciprocal-gamma convolution and coefficient formulas

The coefficient calculation rests on a two-parameter Chu–Vandermonde identity.

**Lemma 4.1** (Asymmetric reciprocal-gamma convolution)**.** *Let $m \in \mathbb{N}_0$ and let $\alpha, \beta > 0$. Then*

$$\sum_{i=0}^{m} \frac{1}{i!(m-i)!\Gamma(\alpha+i)\Gamma(\beta+m-i)} = \frac{(\alpha+\beta+m-1)_m}{m!\Gamma(\alpha+m)\Gamma(\beta+m)}. \tag{4.1}$$

*Both sides extend meromorphically in $(\alpha, \beta)$.*

*Proof.* Multiply the left-hand side by $m!\Gamma(\alpha+m)\Gamma(\beta+m)$. Using

$$\frac{\Gamma(\alpha+m)}{\Gamma(\alpha+i)} = (\alpha+i)_{m-i} = \frac{(\alpha)_m}{(\alpha)_i}$$

and

$$\frac{\Gamma(\beta+m)}{\Gamma(\beta+m-i)} = (\beta+m-i)_i = (-1)^i(1-\beta-m)_i,$$

the resulting sum is

$$(\alpha)_m {}_2F_1\left(\begin{matrix}-m, 1-\beta-m\\ \alpha\end{matrix}; 1\right).$$

Chu–Vandermonde gives $(\alpha+\beta+m-1)_m$. □

Set

$$S_m = S_m(a) = \frac{(2a+m-1)_m}{m!\Gamma(a+m)^2} > 0. \tag{4.2}$$

Taking $\alpha = \beta = a$ in Lemma 4.1 gives

$$[\lambda^m] Z(a,\lambda)^2 = S_m. \tag{4.3}$$

**Theorem 4.2** (Exact coefficient matrices)**.** *There are expansions*

$$A = \sum_{m\ge 0} S_m \alpha_m \lambda^m, \qquad B = \sum_{m\ge 0} S_m \beta_m \lambda^m, \qquad C_\kappa = \sum_{m\ge 0} S_m \gamma_m^{(\kappa)} \lambda^m, \tag{4.4}$$

*where*

$$\alpha_m = \psi_1(a+m), \tag{4.5}$$

$$\beta_0 = 1, \qquad \beta_m = \frac{2a+m-2}{2(a+m-1)} \quad (m \ge 1), \tag{4.6}$$

*and*

$$\gamma_m^{(\kappa)} = 1 + g c_m^{(\kappa)}, \tag{4.7}$$

*with*

$$c_0^{(\kappa)} = 0, \qquad c_1^{(\kappa)} = \frac{\kappa-1}{2}, \tag{4.8}$$

*and, for $m \ge 2$,*

$$c_m^{(\kappa)} = \frac{\kappa m}{2} - \frac{m(2a+m-2)}{2(2a+2m-3)} = \frac{m\left[(\kappa-1)(2a+2m-3)+m-1\right]}{2(2a+2m-3)}. \tag{4.9}$$

*At the sharp endpoint $\kappa = 1$,*

$$c_0 = c_1 = 0, \qquad c_m = \frac{m(m-1)}{2(2a+2m-3)} \quad (m \ge 2). \tag{4.10}$$

*Consequently,*

$$\mathcal{T}(a,\lambda) = \sum_{m=0}^{\infty} S_m M_m \lambda^m, \qquad M_m = \begin{pmatrix}\alpha_m & \sqrt{g}\,\beta_m\\ \sqrt{g}\,\beta_m & 1+gc_m\end{pmatrix}. \tag{4.11}$$

*The theorem does not assert $M_m \succeq 0$ for every $m$: $M_1$ can be indefinite for small $a > 0$ (precisely, on $0 < a < a_*$, where $a_*$ is the unique positive solution of $(4a-1)\psi_1(a) = 4$, numerically $a_* \approx 0.369$; see Section 6). For general $\kappa$, write $M_m^{(\kappa)}$ for the coefficient matrix of $\mathcal{T}_\kappa$, obtained by replacing $1 + gc_m$ with $1 + gc_m^{(\kappa)}$.*

*Proof.* For the first coefficient, define

$$F_m(\delta) = \sum_{i=0}^{m} \frac{1}{i!(m-i)!\Gamma(a+\delta+i)\Gamma(a-\delta+m-i)}.$$

By Lemma 4.1,

$$F_m(\delta) = \frac{(2a+m-1)_m}{m!\Gamma(a+m+\delta)\Gamma(a+m-\delta)}. \tag{4.12}$$

On the one hand,

$$\partial_\delta^2\left[Z(a+\delta,\lambda)Z(a-\delta,\lambda)\right]\big|_{\delta=0} = 2ZZ_{aa} - 2Z_a^2 = -2A. \tag{4.13}$$

On the other hand, $F_m'(0) = 0$ and the logarithmic second derivative of (4.12) is $-2\psi_1(a+m)$ at $\delta = 0$, so

$$F_m''(0) = -2S_m\psi_1(a+m). \tag{4.14}$$

Comparing the coefficient of $\lambda^m$ in (4.13) with (4.14) gives $[\lambda^m]A = S_m\psi_1(a+m)$, proving (4.5).

For $m = 0$, one has $[\lambda^0]B = Z(a,0)^2 = S_0$, so $\beta_0 = 1$. Assume henceforth that $m \geq 1$. Let $I$ be the index of the first factor in $Z(a+\delta,\lambda)Z(a-\delta,\lambda) = \sum_m F_m(\delta)\lambda^m$, so that $[\lambda^m]$ weights $i$ by $w_i(\delta) = \big[i!(m-i)!\Gamma(a+\delta+i)\Gamma(a-\delta+m-i)\big]^{-1}$; for real $\delta$ with $|\delta| < a$, write $\mathbb{E}_\delta$ for expectation under the probability weights $w_i(\delta)/F_m(\delta)$, and put $\mathbb{E} = \mathbb{E}_0$, $\mathsf{D} = m - 2I$, and $J = m - I$. Applying Lemma 4.1 to $\sum_i i\, w_i(\delta)$, with $(\alpha,\beta) = (a+1+\delta, a-\delta)$ and $m-1$ in place of $m$, gives

$$\sum_{i=0}^{m} i\, w_i(\delta) = \frac{(2a+m-1)_{m-1}}{(m-1)!\,\Gamma(a+m+\delta)\Gamma(a+m-1-\delta)}.$$

Dividing by (4.12) yields

$$\mathbb{E}_\delta I = \frac{m(a+m-1-\delta)}{2(a+m-1)},$$

whence

$$\mathbb{E}_\delta \mathsf{D} = \frac{m\delta}{a+m-1}. \tag{4.15}$$

The score $\partial_\delta \log w_i(\delta)\big|_{\delta=0}$ is

$$X = \psi(a+m-I) - \psi(a+I),$$

so differentiating (4.15) at $\delta = 0$ gives $\mathbb{E}(\mathsf{D}X) = \partial_\delta \mathbb{E}_\delta \mathsf{D}|_{\delta=0}$, that is

$$\mathbb{E}(\mathsf{D}X) = \frac{m}{a+m-1}. \tag{4.16}$$

Reading $I = i$ as the $\lambda$-degree carried by the first factor and $J = m-i$ as the degree carried by the second, direct coefficient extraction gives

$$\begin{aligned}
&[\lambda^m]\big(ZZ_{a\Theta_\lambda} - Z_a Z_{\Theta_\lambda}\big) \\
&\quad = \sum_{i=0}^{m} \frac{-J\psi(a+J) + J\psi(a+i)}{i!J!\Gamma(a+i)\Gamma(a+J)} = -S_m\mathbb{E}(JX).
\end{aligned}$$

Together with $[\lambda^m]Z^2 = S_m$, this yields

$$\frac{[\lambda^m]B}{S_m} = 1 - \mathbb{E}(JX) = 1 - \frac{1}{2}\mathbb{E}(\mathsf{D}X),$$

because $J = (m+\mathsf{D})/2$ and $\mathbb{E}X = 0$ by symmetry. This proves (4.6).

For $C_\kappa$, the Euler-derivative terms contribute

$$\kappa\mathbb{E}J - \mathbb{E}J^2 + \mathbb{E}(IJ).$$

Since $I = (m-\mathsf{D})/2$, $J = (m+\mathsf{D})/2$, and $\mathbb{E}\mathsf{D} = 0$ by the symmetry $I \leftrightarrow m-I$,

$$\kappa\mathbb{E}J - \mathbb{E}J^2 + \mathbb{E}(IJ) = \frac{\kappa m}{2} - \frac{1}{2}\mathbb{E}\mathsf{D}^2. \tag{4.17}$$

For $m \geq 2$, cancelling the factors $i(m-i)$ and shifting $j = i-1$ gives

$$\begin{aligned}
S_m(a)\mathbb{E}[I(m-I)] &= \sum_{i=1}^{m-1} \frac{1}{(i-1)!(m-i-1)!\Gamma(a+i)\Gamma(a+m-i)} \\
&= S_{m-2}(a+1).
\end{aligned}$$

Therefore

$$\mathbb{E}[I(m-I)] = \frac{S_{m-2}(a+1)}{S_m(a)} = \frac{m(m-1)(a+m-1)}{2(2a+2m-3)},$$

so

$$\mathbb{E}\mathsf{D}^2 = \frac{m(2a+m-2)}{2a+2m-3}. \tag{4.18}$$

Substitution gives (4.9); the cases $m = 0, 1$ are immediate. □

## 5. Probabilistic interpretation: the finite conditional law

The normalized weights used in the coefficient calculation admit the following finite-law interpretation. Fix $\lambda > 0$ and let $K_1, K_2$ be independent Bessel-law variables with

$$\mathbb{P}(K_j = k) = \frac{\lambda^k}{Z(a,\lambda)\, k!\Gamma(a+k)}, \qquad k \in \mathbb{N}_0.$$

Conditional on $K_1 + K_2 = m$, the common factor $\lambda^m / Z(a,\lambda)^2$ cancels. Thus the conditional law of $I := K_1$ is the symmetric finite law

$$\mathbb{P}(I = i \mid K_1 + K_2 = m) = \frac{1}{S_m} \frac{1}{i!(m-i)!\Gamma(a+i)\Gamma(a+m-i)}, \qquad 0 \le i \le m. \tag{5.1}$$

In the expectations below, the conditioning event $K_1 + K_2 = m$ is understood. Put

$$\mathsf{D} = m - 2I, \qquad X = \psi(a+m-I) - \psi(a+I), \qquad \varrho = \psi_1(a+I) + \psi_1(a+m-I).$$

Symmetry gives $\mathbb{E}\mathsf{D} = \mathbb{E}X = 0$, and the coefficient calculation above gives

$$\alpha_m = \tfrac{1}{2}\mathbb{E}(\varrho - X^2), \qquad \beta_m = 1 - \tfrac{1}{2}\mathbb{E}(\mathsf{D}X), \qquad c_m^{(\kappa)} = \tfrac{\kappa m}{2} - \tfrac{1}{2}\mathbb{E}\mathsf{D}^2. \tag{5.2}$$

For $m \ge 1$, the last two identities are (4.16) and (4.17); for $m = 0$, they are immediate from $I = \mathsf{D} = X = 0$. Differentiating $F_m(\delta)$ twice gives

$$\frac{F_m''(0)}{F_m(0)} = \mathbb{E}(X^2 - \varrho) = -2\psi_1(a+m).$$

Thus, at $\kappa = 1$, the normalized coefficient matrix is

$$N_m := \operatorname{diag}(1, g^{-1/2}) M_m \operatorname{diag}(1, g^{-1/2}) = \begin{pmatrix} \frac{1}{2}\mathbb{E}(\varrho - X^2) & 1 - \frac{1}{2}\mathbb{E}(\mathsf{D}X) \\ 1 - \frac{1}{2}\mathbb{E}(\mathsf{D}X) & g^{-1} + \frac{m}{2} - \frac{1}{2}\mathbb{E}\mathsf{D}^2 \end{pmatrix}. \tag{5.3}$$

This probabilistic reformulation is not needed for the subsequent Gram proof; it records the natural conditional origin of the finite weights and packages the coefficient identities in (5.2). The probabilistic interpretation is restricted to $a > 0$; only the resulting algebraic coefficient identities are subsequently continued beyond that range.

## 6. Explicit Gram representations

We require elementary two-sided trigamma estimates. The integral representation

$$\psi_1(y) = \int_0^\infty e^{-y\tau} \frac{\tau}{1 - e^{-\tau}}\, d\tau, \qquad y > 0, \tag{6.1}$$

and the series $\psi_1(y) = \sum_{r \ge 0} (y+r)^{-2}$ are standard [13].

**Lemma 6.1** (Trigamma bounds)**.** *For every $y > 0$,*

$$\psi_1(y) > \frac{1}{y} + \frac{1}{2y^2} > \frac{1}{y}. \tag{6.2}$$

*For $y > 1/2$,*

$$\psi_1(y) < \frac{1}{y - 1/2}. \tag{6.3}$$

*Consequently,*

$$\frac{1}{\psi_1(a)} > a - \frac{1}{2} \qquad (a > 0), \tag{6.4}$$

*where the assertion is immediate for $a \le 1/2$.*

*Proof.* For $f(x) = (y+x)^{-2}$, strict convexity on every interval $[r, r+1]$ gives the strict trapezoidal inequality

$$\int_r^{r+1} f(x)\,\mathrm{d}x < \frac{f(r) + f(r+1)}{2}.$$

Summing and passing to the limit yields

$$\psi_1(y) = \sum_{r\geq 0} f(r) > \int_0^\infty f(x)\,\mathrm{d}x + \frac{1}{2}f(0) = \frac{1}{y} + \frac{1}{2y^2},$$

which proves (6.2). For the upper bound, the elementary inequality $\tau < 2\sinh(\tau/2)$ for $\tau > 0$ is equivalent to $\tau/(1-e^{-\tau}) < e^{\tau/2}$. Substitution in (6.1) gives (6.3). Taking reciprocals proves (6.4) when $a > 1/2$. □

Fix $m \geq 1$ and write

$$x = a + m, \qquad q = a + \frac{m}{2} - 1. \tag{6.5}$$

In $\ell^2(\mathbb{N}_0)$ define

$$u_r^{(m)} = \frac{1}{x+r}, \qquad v_r^{(m)} = \frac{q}{x+r-1}, \qquad r \geq 0. \tag{6.6}$$

Then

$$\|u^{(m)}\|^2 = \psi_1(x) = \alpha_m, \tag{6.7}$$

$$\langle u^{(m)}, v^{(m)}\rangle = q\sum_{r\geq 0}\frac{1}{(x+r-1)(x+r)} = \frac{q}{x-1} = \beta_m, \tag{6.8}$$

and

$$\|v^{(m)}\|^2 = q^2\psi_1(x-1). \tag{6.9}$$

Define

$$\rho_m(a) = \frac{1}{g} + c_m - q^2\psi_1(x-1). \tag{6.10}$$

Call the coefficient matrix $M_m$ *stable* when $m \geq 2$, or $m = 1$ and $a \geq 1/2$. For vectors $x, y$ in a Hilbert space, $\mathrm{Gram}(x, y)$ denotes the symmetric $2 \times 2$ matrix with entries $\langle x, x\rangle$, $\langle x, y\rangle$, $\langle y, y\rangle$.

**Theorem 6.2** (Gram representation of stable coefficients)**.** *If either $m \geq 2$ and $a > 0$, or $m = 1$ and $a \geq 1/2$, then $\rho_m(a) > 0$. In $\ell^2(\mathbb{N}_0) \oplus \mathbb{R}$, put*

$$\xi_m = (u^{(m)}, 0), \qquad \eta_m = (v^{(m)}, \sqrt{\rho_m(a)}).$$

*Then*

$$N_m = \mathrm{Gram}(\xi_m, \eta_m), \tag{6.11}$$

$$M_m = \mathrm{Gram}(\xi_m, \sqrt{g}\,\eta_m). \tag{6.12}$$

*In particular, $M_m$ is positive definite in these ranges.*

*Proof.* Assume first that $m \geq 2$. Then $x - 1 > 1/2$, so Lemma 6.1 gives

$$\rho_m(a) > a - \frac{1}{2} + c_m - \frac{q^2}{x - 3/2}.$$

At the sharp endpoint the right-hand side simplifies exactly to

$$a - \frac{1}{2} + \frac{m(m-1)}{2(2a+2m-3)} - \frac{(a+m/2-1)^2}{a+m-3/2} = \frac{m-1}{2(2a+2m-3)} > 0. \tag{6.13}$$

For $m = 1$, $q = a - 1/2$ and $c_1 = 0$. If $a > 1/2$, then

$$q^2\psi_1(a) < q < \frac{1}{\psi_1(a)},$$

by (6.3) and (6.4). Hence $\rho_1(a) > 0$. At $a = 1/2$, $q = 0$ and the conclusion is immediate. The Gram identities now follow from (6.7)–(6.10). $\square$

The constant matrix is

$$M_0 = \begin{pmatrix} g & \sqrt{g} \\ \sqrt{g} & 1 \end{pmatrix} = \begin{pmatrix} \sqrt{g} \\ 1 \end{pmatrix} \begin{pmatrix} \sqrt{g} & 1 \end{pmatrix}, \tag{6.14}$$

a rank-one positive-semidefinite matrix. The possible anomaly at $m = 1$ is genuine: using $\psi_1(a+1) = g - a^{-2}$ and $\beta_1 = (2a-1)/(2a)$,

$$\det M_1 = \psi_1(a+1) - g\left(\frac{2a-1}{2a}\right)^2 = \frac{(4a-1)g-4}{4a^2}. \tag{6.15}$$

Set $f(a) = (4a-1)\psi_1(a) - 4$, so that $\det M_1$ has the sign of $f(a)$. Since $\psi_1(a) = \sum_{r\ge 0}(a+r)^{-2}$, for $0 < a < 1/2$

$$f'(a) = 2\sum_{r\ge 0} \frac{2r+1-2a}{(a+r)^3} > 0,$$

while $f(a) \to -\infty$ as $a \downarrow 0$ and $f(1/2) = \psi_1(1/2) - 4 = \pi^2/2 - 4 > 0$. Hence $f$ has a unique zero $a_* \in (0, 1/2)$; numerically, $a_* = 0.3690738484\ldots$. Combining this with Theorem 6.2 for $a \ge 1/2$, and using the positivity of the diagonal entries of $M_1$, we conclude that $M_1$ is indefinite for $0 < a < a_*$, positive semidefinite of rank one at $a = a_*$, and positive definite for $a > a_*$. Thus $M_1$ is the only coefficient matrix that can be indefinite at the endpoint $\kappa = 1$.

## 7. Mixed determinants and coefficientwise positivity

For symmetric $2 \times 2$ matrices $X = (x_{ij})$ and $Y = (y_{ij})$, define

$$\mathrm{MD}(X, Y) = x_{11}y_{22} + x_{22}y_{11} - 2x_{12}y_{12}. \tag{7.1}$$

This is the polarization of the determinant.

**Lemma 7.1** (Wedge-square positivity)**.** *If* $X, Y \succeq 0$, *then* $\mathrm{MD}(X, Y) \ge 0$. *If* $X \ne 0$ *and* $Y \succ 0$, *then* $\mathrm{MD}(X, Y) > 0$.

*Proof.* Write $X = \sum_r x_r x_r^T$ and $Y = \sum_s y_s y_s^T$. Then

$$\mathrm{MD}(X, Y) = \sum_{r,s} (x_{r,1}y_{s,2} - x_{r,2}y_{s,1})^2.$$

If this sum vanished and $X \ne 0$, fix a nonzero Gram vector $x_r$. Every Gram vector $y_s$ would then be parallel to $x_r$, forcing $Y$ to have rank at most one, contrary to $Y \succ 0$. $\square$

Expanding the determinant of (4.11) gives

$$\Delta_n = \frac{1}{2}\sum_{k=0}^{n} S_k S_{n-k}\,\mathrm{MD}(M_k, M_{n-k}). \tag{7.2}$$

If $a \ge 1/2$, every $M_m$ is positive semidefinite and every $M_m$ with $m \ge 1$ is positive definite. Thus every summand in (7.2) is nonnegative, while the pair $(M_0, M_n)$ is strictly positive by Lemma 7.1, because $M_0 \ne 0$ and $M_n \succ 0$. Hence $\Delta_n > 0$.

Now assume $0 < a < 1/2$. By (4.6), $\beta_1 < 0$ while $\beta_m > 0$ for $m \ge 2$. Since every diagonal entry is positive, for $m \ge 2$,

$$\mathrm{MD}(M_1, M_m) = \alpha_1(1 + gc_m) + \alpha_m - 2g\beta_1\beta_m > 0. \tag{7.3}$$

The linear coefficient is also positive. Indeed,

$$\mathrm{MD}(M_0, M_1) = \frac{ag-1}{a^2} > 0, \tag{7.4}$$

and hence

$$\Delta_1 = S_0 S_1\,\mathrm{MD}(M_0, M_1) = \frac{2(ag-1)}{a^3\Gamma(a)^4} > 0. \tag{7.5}$$

For $n \geq 3$, classify the pairs $(k, n-k)$ in (7.2). If neither index is 1, Gram positivity applies; if one index is 1, the other is at least 2 and (7.3) applies. Thus every summand is nonnegative, and the pair $(M_0, M_n)$ is strictly positive. The indefiniteness of $M_1$ can therefore contribute a potentially negative summand only through its self-pairing $\mathrm{MD}(M_1, M_1) = 2\det M_1$ in $\Delta_2$.

### 7.1. **The exceptional degree-two coefficient.**

**Lemma 7.2** (Exceptional degree-two decomposition)**.** *For every $a > 0$, one has $\Delta_2(a) > 0$. More precisely, let*

$$t = \psi_1(a+1), \qquad \mathcal{R}_a = t - \frac{1}{a+1}. \tag{7.6}$$

*Then*

$$\Delta_2 = \frac{Q_*(a,t)}{2a^6(a+1)^3\Gamma(a)^4}, \tag{7.7}$$

*where*

$$\begin{aligned} Q_*(a,t) = {} & 2a^4(a+1)^2\mathcal{R}_a^2 \\ & + 2a^2(a+1)^2(8a^2+3a+1)\mathcal{R}_a \\ & + 2a(a+1)(5a+3). \end{aligned} \tag{7.8}$$

*Proof.* From (7.2),

$$\Delta_2 = S_0 S_2 \,\mathrm{MD}(M_0, M_2) + S_1^2 \det M_1. \tag{7.9}$$

Here

$$S_0 = \frac{1}{\Gamma(a)^2}, \qquad S_1 = \frac{2}{a\Gamma(a)^2}, \qquad S_2 = \frac{2a+1}{a^2(a+1)\Gamma(a)^2},$$

and, recalling $\det M_1 = t - g((2a-1)/(2a))^2$ from (6.15),

$$\mathrm{MD}(M_0, M_2) = g\left(1 + \frac{g}{2a+1}\right) + \psi_1(a+2) - 2g\frac{a}{a+1}.$$

Substituting $g = t + a^{-2}$ and $\psi_1(a+2) = t - (a+1)^{-2}$ into (7.9) and collecting over the denominator in (7.7) first gives

$$\begin{aligned} Q_*(a,t) = {} & 2a^4(a+1)^2t^2 \\ & + 2a^2(8a^4+17a^3+13a^2+5a+1)t \\ & + 2a(-8a^4-10a^3+a^2+7a+3). \end{aligned}$$

After multiplication by the common denominator in (7.7), the preceding formula is a polynomial identity in $(a,t)$. Replacing $t$ by $\mathcal{R}_a + (a+1)^{-1}$ and collecting this polynomial yields the identity (7.8).

By (6.1),

$$\mathcal{R}_a = \int_0^\infty e^{-(a+1)s}\left(\frac{s}{1-e^{-s}} - 1\right) \mathrm{d}s > 0. \tag{7.10}$$

All three coefficients in (7.8) are positive for $a > 0$, so $\Delta_2 > 0$.

Equivalently, let

$$f_a(s) = e^{-(a+1)s/2}\sqrt{\frac{s}{1-e^{-s}} - 1}, \qquad \|f_a\|_2^2 = \mathcal{R}_a,$$

and define

$$\mathcal{H}_a = \left(L^2(0,\infty)\widehat{\otimes}L^2(0,\infty)\right) \oplus L^2(0,\infty) \oplus \mathbb{R}.$$

Here $\widehat{\otimes}$ denotes the Hilbert-space tensor product. Then

$$Q_* = \left\|\sqrt{2}a^2(a+1)f_a \otimes f_a \oplus \sqrt{2}a(a+1)\sqrt{8a^2+3a+1}\, f_a \oplus \sqrt{2a(a+1)(5a+3)}\right\|_{\mathcal{H}_a}^2. \tag{7.11}$$

□

*Proof of Theorem 2.1.* The preceding argument, together with Lemma 7.2, proves the endpoint assertion $\Delta_n^{(1)}(a) > 0$ for every $a > 0$ and $n \geq 1$. Since $A$ and $Z$ have positive coefficients and

$$[\lambda^n]Z_{\Theta_\lambda} = \frac{n}{n!\Gamma(a+n)} > 0 \qquad (n \geq 1),$$

every positive-degree coefficient of $AZZ_{\Theta_\lambda}$ is strictly positive: in degree $n$, for example, the product of the constant coefficients of $A$ and $Z$ with $[\lambda^n]Z_{\Theta_\lambda}$ is positive. Therefore

$$\Delta^{(\kappa)} = \Delta^{(1)} + g(\kappa - 1)A\,ZZ_{\Theta_\lambda} \tag{7.12}$$

shows that all coefficients remain strictly positive for $\kappa > 1$.

For the converse, the general $m = 1$ coefficient gives

$$\mathrm{MD}(M_0, M_1^{(\kappa)}) = \frac{\frac{\kappa-1}{2}(ag)^2 + ag - 1}{a^2}. \tag{7.13}$$

If $\kappa < 1$, then $ag \to +\infty$ as $a \downarrow 0$, so (7.13), and hence $\Delta_1^{(\kappa)}$, is negative for all sufficiently small $a > 0$. This proves the sharp classification.

Finally, $\Delta(a, \cdot)$ is entire, so each differentiated Maclaurin series converges locally uniformly. Hence, for every $r \geq 0$ and $\lambda > 0$,

$$\partial_\lambda^r \Delta(a, \lambda) = \sum_{n \geq \max\{1,r\}} \frac{n!}{(n-r)!}\Delta_n(a)\lambda^{n-r} > 0.$$

Thus $\Delta(a, \cdot)$ is strictly absolutely monotone. Also $A > 0$ for $\lambda > 0$ by (4.4) and (4.5). Hence $\Delta > 0$ and the leading principal entry is positive, so $\mathcal{T} \succ 0$. □

## 8. Bessel consequences and sharpness

The coefficientwise theorem of Section 7, transported through the exact reduction of Section 3, gives the sharp Bessel inequality; the same computation fixes its optimal constants, and its continuation past the positive-series domain exhibits the negative-order failure.

*Proof of Theorem 2.3 and Corollary 2.4.* The congruence (3.5) and Theorem 2.1 give positive definiteness for $z > 0$, and (3.6) gives the determinant inequality. As $z \downarrow 0$, equivalently $\lambda \downarrow 0$, the power series for $L = \log Z$ gives $L_{\Theta_\lambda}, L_{a\Theta_\lambda}, L_{\Theta_\lambda\Theta_\lambda} \to 0$. Hence (3.2)–(3.3) imply

$$G_\nu \longrightarrow g, \qquad 1 + P_\nu \longrightarrow 2, \qquad H_\nu + \frac{4}{g} \longrightarrow \frac{4}{g},$$

which is the matrix limit stated in Corollary 2.4. □

*Proof of Proposition 2.5.* The exact minimality of the $z$-independent correction follows from the argument after (3.6). The "if" direction for the deformation follows from (3.7) and Theorem 2.1; the "only if" direction follows from the negative leading term in (3.9) for every $\kappa < 1$ and all sufficiently small $a > 0$. □

The behavior changes qualitatively once the order crosses the endpoint $\nu = -1$ of the positive-series domain.

**Definition 8.1** (Negative-order convention)**.** Let $\nu \leq -1$ be real. On each connected component of $\{z > 0 : I_\nu(z) \neq 0\}$, define $G_\nu$, $P_\nu$, and $H_\nu$ by (2.8)–(2.10) with $\log I_\nu$ replaced by $\log |I_\nu|$. For nonintegral $\nu$, define $D_\nu$ by (2.11). At a negative integer $\nu = -n$, the convention means explicitly

$$\begin{aligned} G_{-n}(z) &= \left(\frac{\partial_\nu I_\nu(z)|_{\nu=-n}}{I_n(z)}\right)^2 - \frac{\partial_\nu^2 I_\nu(z)\big|_{\nu=-n}}{I_n(z)}, \\ P_{-n}(z) &= \Theta_z\left(\frac{\partial_\nu I_\nu(z)|_{\nu=-n}}{I_n(z)}\right), \\ H_{-n}(z) &= 2\big(\Theta_z \log I_n(z) + n\big) - \Theta_z^2 \log I_n(z). \end{aligned}$$

Thus the order derivatives are taken before setting $\nu = -n$. We interpret $1/\psi_1(\nu+1)$ by its local analytic extension, whose value at the pole is 0, and define

$$D_{-n}(z) = G_{-n}(z)H_{-n}(z) - \big(1 + P_{-n}(z)\big)^2.$$

On a component where $I_\nu > 0$, this convention agrees with the ordinary real logarithm.

For every real $a \notin \{0, -1, -2, \ldots\}$, the partial-fraction expansion

$$\psi_1(a) = \sum_{k=0}^{\infty} \frac{1}{(a+k)^2} \tag{8.1}$$

holds and shows that $g = \psi_1(a) > 0$ [13]. Hence the positive real square root $\sqrt{g}$ occurring in (2.5) and (3.5) remains well defined on every real nonpole component.

**Lemma 8.2** (Continuation beyond the positive-series domain)**.** *For each fixed coefficient index, the scalar coefficient identities in* (4.4)*–*(4.9)*, and every resulting finite-convolution formula for a coefficient of* $\Delta$*, extend meromorphically in* $a$*. This assertion concerns the scalar entries and* $\Delta$*; it does not require a single-valued meromorphic choice of* $\sqrt{\psi_1(a)}$ *on the complex plane. If* $a \in \mathbb{R} \setminus \{0, -1, -2, \ldots\}$*,* $\lambda > 0$*, and* $Z(a, \lambda) \neq 0$*, then* (3.4)*,* (3.5)*, and* (3.6) *remain valid, with the positive real square root of* $\psi_1(a)$ *and with logarithmic derivatives interpreted through* $\log |Z|$ *and* $\log |I_{a-1}|$ *on the relevant nonvanishing component.*

*Proof.* Each summand of (2.1) is entire in $a$. Uniform Stirling estimates for $a$ in compact sets, together with Cauchy estimates for the derivatives of $1/\Gamma$, give locally uniform convergence of the series and of every fixed-order parameter and Euler derivative on compact subsets of $\mathbb{C}^2$; hence $Z$ and these derivatives are entire in $(a, \lambda)$. For each fixed coefficient index, both sides of each scalar coefficient identity are meromorphic functions of the single variable $a$ and agree on $a > 0$; the one-variable identity theorem gives the continuation. Each coefficient of $\Delta = AC - gB^2$ is a finite convolution of such scalar coefficient functions and is therefore meromorphic as well.

On the real nonpole set, (8.1) gives $g = \psi_1(a) > 0$, so the positive real square root used in the matrix formulas is well defined. On each connected component of the real nonvanishing set, $Z$ has constant sign, so $\partial_a \log |Z| = Z_a/Z$ and the corresponding mixed and second derivative identities hold there. Finally, (3.1) gives $\log |I_{a-1}(2\sqrt{\lambda})| = \frac{a-1}{2} \log \lambda + \log |Z(a, \lambda)|$ where $Z \neq 0$, leaving the derivations of (3.4), (3.5), and (3.6) unchanged on that component. $\square$

No negative-order obstruction is visible in the leading $z^2$-term away from the poles. The positivity needed below follows from (8.1).

**Proposition 8.3** (Small-argument expansion away from the poles)**.** *Let* $a = \nu + 1 \in \mathbb{R} \setminus \{0, -1, -2, \ldots\}$*. On the nonvanishing component of* $I_\nu$ *adjacent to* $z = 0$*, and for this fixed* $a$*,*

$$D_\nu(z) = \frac{2\big(a\psi_1(a) - 1\big)}{a^3\psi_1(a)} z^2 + O(z^4) \qquad (z \downarrow 0). \tag{8.2}$$

*The coefficient is positive for every such real* $a$*, positive or negative.*

*Proof.* For fixed $\nu \notin \{-1, -2, -3, \ldots\}$,

$$I_\nu(z) = \frac{(z/2)^\nu}{\Gamma(\nu+1)}\big(1 + O(z^2)\big) \qquad (z \downarrow 0),$$

so $I_\nu$ is nonzero with constant sign for all sufficiently small $z > 0$; hence the stated component exists. By Lemma 8.2, (7.5) and (3.6) apply throughout the stated $a$-domain. Substituting (7.5), with $Z(a, 0) = 1/\Gamma(a)$ and $\lambda = z^2/4$, gives (8.2). For $a > 0$, (6.2) gives $a\psi_1(a) > 1$. For negative nonintegral $a$, (8.1) gives $\psi_1(a) > 0$, so $a\psi_1(a) - 1 < 0$ and $a^3\psi_1(a) < 0$ and their ratio is positive. $\square$

Although the first local term is positive, coefficientwise positivity of the continued determinant $\Delta$ already fails on the next interval on which $1/\Gamma(a) > 0$. Indeed, for $-3 < \nu < -2$ one has $a = \nu + 1 \in (-2, -1)$ and $\Gamma(a) > 0$, so the constant coefficient $1/\Gamma(a)$ of $Z(a, \lambda)$ is positive.

**Proposition 8.4** (Degree-two obstruction beyond the positive-series domain)**.** *If $-2 < a < -1$, equivalently $-3 < \nu < -2$, then*

$$[\lambda^2]\Delta(a, \lambda) = \Delta_2(a) < 0. \tag{8.3}$$

*Thus the coefficientwise positivity theorem for $\Delta$ cannot be extended to that interval.*

*Proof.* By Lemma 8.2, the identity (7.7)–(7.8) persists on $-2 < a < -1$. Set $b = -a \in (1, 2)$. Since $a + 1 \in (-1, 0)$, (8.1) gives $t = \psi_1(a + 1) > 0$,

$$\mathcal{R}_a = t - \frac{1}{a+1} > \frac{1}{b-1}.$$

The first term in (7.8) is positive. The second and third terms satisfy

$$\begin{aligned} &2b^2(b-1)^2(8b^2 - 3b + 1)\mathcal{R}_a - 2b(b-1)(5b-3) \\ &\quad > 2b(b-1)\big[8b^3 - 3b^2 - 4b + 3\big] > 0. \end{aligned}$$

Indeed, the polynomial in brackets equals 4 at $b = 1$ and has derivative $24b^2 - 6b - 4 > 0$ for $b \geq 1$. Hence $Q_*(a, t) > 0$. But $2a^6(a+1)^3\Gamma(a)^4 < 0$, proving (8.3). ☐

There is also genuine pointwise failure. It is detected at the negative integer orders, not by (8.2).

**Lemma 8.5** (Order derivatives at a negative integer)**.** *Let $n \geq 2$ be an integer and set*

$$b_n = (-1)^{n+1}2^{2n}n!(n-1)!.$$

*Then, as $z \downarrow 0$,*

$$\frac{\partial_\nu I_\nu(z)|_{\nu=-n}}{I_n(z)} = b_n z^{-2n}\big(1 + O(z^2)\big), \tag{8.4}$$

$$\Theta_z\left(\frac{\partial_\nu I_\nu(z)|_{\nu=-n}}{I_n(z)}\right) = -2nb_n z^{-2n}\big(1 + O(z^2)\big), \tag{8.5}$$

$$\frac{\partial_\nu^2 I_\nu(z)|_{\nu=-n}}{I_n(z)} = O\big(z^{-2n}(1 + |\log z|)\big). \tag{8.6}$$

*Proof.* The connection formula [12, Eq. 10.27.4] and its first-order derivative at $\nu = -n$ give

$$I_\nu(z) = I_{-\nu}(z) - \frac{2}{\pi}\sin(\pi\nu)K_\nu(z), \tag{8.7}$$

$$\partial_\nu I_\nu(z)\big|_{\nu=-n} = -\partial_\alpha I_\alpha(z)\big|_{\alpha=n} - 2(-1)^n K_n(z). \tag{8.8}$$

The convergent power and power–log expansions at the origin [12, Secs. 10.27, 10.31, and 10.38] imply

$$\begin{aligned} I_n(z) &= \frac{z^n}{2^n n!}\big(1 + O(z^2)\big), & \Theta_z I_n(z) &= \frac{nz^n}{2^n n!}\big(1 + O(z^2)\big), \\ K_n(z) &= 2^{n-1}(n-1)!z^{-n}\big(1 + O(z^2)\big), & \Theta_z K_n(z) &= -n2^{n-1}(n-1)!z^{-n}\big(1 + O(z^2)\big). \end{aligned}$$

For $\alpha$ near $n$, the series for $I_\alpha(z)$ and its first two $\alpha$-derivatives and one application of $\Theta_z$ converge uniformly for small $z > 0$, since $\partial_\alpha^j\big[1/\Gamma(\alpha + k + 1)\big]$ grows at most polynomial–logarithmically against the factorial denominators. Hence, differentiating the prefactor $(z/2)^\alpha$,

$$\partial_\alpha I_\alpha(z)|_{\alpha=n}, \quad \Theta_z\partial_\alpha I_\alpha(z)|_{\alpha=n} = O\big(z^n(1 + |\log z|)\big),$$

$$\partial_\alpha^2 I_\alpha(z)|_{\alpha=n} = O\big(z^n(1 + |\log z|)^2\big). \tag{8.9}$$

Substitution into (8.8) proves (8.4). Applying the quotient rule before taking the asymptotic expansion proves (8.5). Here the differentiated power–log remainders remain of the displayed order; moreover $z^{2n}|\log z| = O(z^2)$ for $n \geq 2$.

Differentiating (8.7) twice, using $K_{-\nu} = K_\nu$, gives

$$\partial_\nu^2 I_\nu(z)\big|_{\nu=-n} = \partial_\alpha^2 I_\alpha(z)\big|_{\alpha=n} + 4(-1)^n\partial_\alpha K_\alpha(z)\big|_{\alpha=n}. \tag{8.10}$$

The estimate (8.9) handles the $I_\alpha$ term. The exact integer-order identity [14, Eq. 10.38.4]

$$\partial_\alpha K_\alpha(z)\big|_{\alpha=n} = \frac{n!}{2(z/2)^n}\sum_{k=0}^{n-1}\frac{(z/2)^k K_k(z)}{k!(n-k)}$$

and the expansions of $K_0,\dots,K_{n-1}$ give $\partial_\alpha K_\alpha|_{\alpha=n} = O(z^{-n}(1+|\log z|))$. Division by $I_n(z) \asymp z^n$ proves (8.6). ☐

**Proposition 8.6** (Pole-limit pointwise obstruction)**.** *Under the convention of Definition 8.1, for every integer $n \ge 2$,*

$$D_{-n}(z) = -4n(n-1)2^{4n}\big[n!(n-1)!\big]^2 z^{-4n}\big(1+o(1)\big) \qquad (z \downarrow 0). \tag{8.11}$$

*Consequently, for every $n \ge 2$ there exist $z_n > 0$ and $\varepsilon_n > 0$ such that*

$$I_\nu(z_n) > 0 \quad \text{and} \quad D_\nu(z_n) < 0 \qquad (|\nu+n| < \varepsilon_n).$$

*In particular, there exist $z_0 > 0$ and $\varepsilon > 0$ such that*

$$D_\nu(z_0) < 0 \qquad (-2-\varepsilon < \nu < -2).$$

*For every $\nu \in (-3,-2)$ one has $I_\nu(z) > 0$ for all $z > 0$, so $\{z > 0 : I_\nu(z) \ne 0\} = (0,\infty)$ and these counterexamples occur inside that global positive component.*

*Proof.* First, if $a$ is near the pole $1-n$, then

$$\psi_1(a) = \frac{1}{(a+n-1)^2} + h_n(a),$$

where $h_n$ is analytic near $1-n$. It follows that $1/\psi_1(a)$ extends analytically across the pole, with a double zero there. Thus the reciprocal-trigamma correction in $D_{-n}$ is exactly zero.

Let $\ell_n(z) = \partial_\nu \log I_\nu(z)\big|_{\nu=-n}$. Since $I_{-n} = I_n$, Lemma 8.5 gives

$$G_{-n}(z) = \ell_n(z)^2 - \frac{\partial_\nu^2 I_\nu(z)|_{\nu=-n}}{I_n(z)} = b_n^2 z^{-4n}\big(1+o(1)\big), \tag{8.12}$$

$$P_{-n}(z) = \Theta_z \ell_n(z) = -2nb_n z^{-2n}\big(1+O(z^2)\big). \tag{8.13}$$

Finally, since $I_{-n} = I_n$ and $I_n(z) = (z^n/2^n n!)(1+O(z^2))$, one has $\Theta_z \log I_n \to n$ and $\Theta_z^2 \log I_n \to 0$ as $z \downarrow 0$, so

$$H_{-n}(z) = 2\big(\Theta_z \log I_n + n\big) - \Theta_z^2 \log I_n = 4n + o(1). \tag{8.14}$$

Using (8.12)–(8.14) and the vanishing reciprocal-trigamma correction,

$$\begin{aligned} D_{-n}(z) &= G_{-n}(z)H_{-n}(z) - \big(1+P_{-n}(z)\big)^2 \\ &= \big(4n-4n^2\big)\, b_n^2 z^{-4n}\big(1+o(1)\big), \end{aligned}$$

which is (8.11) because $b_n^2 = 2^{4n}[n!(n-1)!]^2$.

Choose $z_n > 0$ sufficiently small that $D_{-n}(z_n) < 0$. For fixed $z_n$, the function $I_\nu(z_n)$ is entire in $\nu$ and $I_{-n}(z_n) = I_n(z_n) > 0$. Hence there is a neighborhood of $-n$ on which $I_\nu(z_n) \ne 0$. On that neighborhood, all quotients and order derivatives entering $G_\nu(z_n)$, $P_\nu(z_n)$, and $H_\nu(z_n)$ are analytic in $\nu$, and the reciprocal-trigamma term has the analytic extension established above. Thus $D_\nu(z_n)$ is continuous there. Shrinking the neighborhood if necessary gives $I_\nu(z_n) > 0$ and $D_\nu(z_n) < 0$ for $|\nu+n| < \varepsilon_n$ with some $\varepsilon_n > 0$.

Taking $n = 2$ and restricting to the left-hand neighborhood gives the stated interval adjacent to $-2$. Finally, if $-3 < \nu < -2$ and $s = -\nu \in (2,3)$, then (8.7) reads

$$I_\nu(z) = I_s(z) + \frac{2}{\pi}\sin(\pi s)K_s(z) > 0 \qquad (z > 0),$$

because $I_s$, $K_s$, and $\sin(\pi s)$ are positive. ☐

**Remark 8.7.** The sharp theorem therefore has three distinct negative-order features: positive leading small-$z$ behavior away from poles, failure of coefficientwise positivity of $\Delta$ on all of $-3 < \nu < -2$, and actual pointwise failure of the Bessel inequality in neighborhoods of every negative integer $-n$, $n \ge 2$.

## 9. Comparison with related positivity results

The scalar quantity $A = Z_a^2 - ZZ_{aa}$ is the infinitesimal symmetric-shift Turánian of the reciprocal-gamma series. Indeed, $Z(a+\delta,\lambda)Z(a-\delta,\lambda) - Z(a,\lambda)^2 = -A(a,\lambda)\delta^2 + O(\delta^4)$. The discrete Wright log-concavity and coefficientwise Turán results of Kalmykov and Karp specialize to this reciprocal-gamma series and yield the scalar coefficient sign of $A$ [8, 11]. Their gamma-ratio results similarly treat scalar two-function Turánians under parameter shifts [9, 10].

The additional quantities $B$ and $C$ are not scalar parameter-shift Turánians: $B$ contains a mixed parameter–Euler derivative, while $C$ contains the sharp argument-curvature expression and the trigamma normalization that becomes the least $z$-independent correction under the Bessel congruence. The result proved here is the coefficientwise positivity of the coupled determinant $AC - \psi_1(a)B^2$, even though the coefficient matrix $M_1$ can be indefinite. We do not know a formal deduction of this determinant statement from the scalar log-concavity theorems cited above.

Classical Bessel Turán inequalities compare neighboring orders, for example $I_\nu^2 - I_{\nu-1}I_{\nu+1}$, or control ratios and logarithmic derivatives [1, 2, 4, 18]; higher-order Turán and Hankel determinants in the order alone have also been studied [3]. In contrast, (2.14) couples second curvature in the order with first mixed order–argument curvature and second argument curvature.

Recent work of Salazar proves that the kernel $(x,s) \mapsto I_s(x)$ is strictly totally positive for $x > 0$ and $s \geq 0$ [15], extending integer-order results of Buchstaber and Glutsyuk [5]; a companion moment-transport framework builds all-order Hankel and Turán determinant hierarchies for special functions from positive moment representations [16]. Their determinant hierarchies are respectively of evaluation-minor and Hankel-moment type; none of these papers states or proves the coefficientwise mixed-derivative determinant studied here. To compare the ordinary confluent minors of $I_s(x)$ with our inequality, write

$$F(s,y) = \log I_s(e^y), \qquad p = F_{sy}, \qquad G = -F_{ss}.$$

On the domain $s \geq 0$ of that theorem, the ordinary confluent minors of the kernel are nonnegative: the Vandermonde-normalized confluent limits of orders two and three give

$$p \geq 0 \qquad \text{and} \qquad p + sG_y \geq 0$$

(Proposition A.1). These are related differential inequalities, but these sign inequalities alone do not yield the Bessel–Schur inequality proved here.

Indeed, the signs of all total-positivity minors, including their confluent limits, are preserved under positive row and column rescalings of the kernel, whereas our Schur determinant is not invariant under such rescalings. To make the obstruction explicit, replace $I_s(e^y)$ by $e^{cy}I_s(e^y)$ and write $\widetilde{F} = F + cy$. Then

$$\widetilde{G} = -\widetilde{F}_{ss} = G, \qquad \widetilde{p} = \widetilde{F}_{sy} = p, \qquad \widetilde{H} = 2(\widetilde{F}_y - s) - \widetilde{F}_{yy} = H + 2c.$$

Consequently, writing the Schur determinant as

$$D_s(e^y) = G\left(H + \frac{4}{\psi_1(s+1)}\right) - (1+p)^2,$$

the rescaled kernel has

$$\widetilde{D}_s(e^y) = D_s(e^y) + 2cG,$$

although all total-positivity signs are unchanged.

There is also a domain mismatch. The standard Hankel expansion, locally uniformly for complex $s$ in a neighborhood of each compact real $s$-interval, gives [12, Secs. 10.40(i), 10.40(iii)]

$$I_s(x) = \frac{e^x}{\sqrt{2\pi x}}\left(1 - \frac{4s^2-1}{8x} + O(x^{-2})\right), \qquad x \to \infty. \tag{9.1}$$

On any such fixed complex neighborhood, the parenthesized factor is uniformly close to 1 and hence nonzero for all sufficiently large $x$. It therefore has the analytic logarithm selected by the branch near 1, and (9.1) yields

$$\log I_s(x) = x - \frac{1}{2}\log(2\pi x) - \frac{4s^2-1}{8x} + O(x^{-2}).$$

The $x$-differentiated Hankel expansion gives the corresponding locally uniform estimate after applying $x\partial_x$. Cauchy's integral formula in the order variable then permits differentiation of these remainders with respect to $s$, whence

$$\partial_s\big(x\partial_x \log I_s(x)\big) = \frac{s}{x} + O(x^{-2}), \qquad x \to \infty.$$

Thus the kernel fails even local total positivity of order two for negative $s$ and sufficiently large $x$, while the present inequality holds throughout $-1 < \nu < 0$. The inequality therefore cannot be deduced from the gauge-invariant sign data of the existing total-positivity theorem or its ordinary confluent minors alone. Any derivation from total positivity would require additional Bessel-specific input, such as a canonical endpoint normalization or a bordered nonlocal construction.

The reciprocal-gamma normalization $Z(a,\lambda)$ is the partition function of the Bessel law $\mathbb{P}(K = k) \propto \lambda^k/(k!\Gamma(a+k))$, whose randomized-gamma representation, simulation, and relation to the von Mises–Fisher family were studied by Yuan and Kalbfleisch [20]; see also Devroye [7]. Bessel laws also arise as complete conditionals in Poisson–Gamma latent models [17].

## 10. Further questions

**Question 10.1** (Wright generalization)**.** For $\rho > 0$, let

$$Z_\rho(a,\lambda) = \sum_{k\ge 0} \frac{\lambda^k}{k!\Gamma(a+\rho k)}, \qquad L_\rho = \log Z_\rho,$$

and, for $\kappa \in \mathbb{R}$, define

$$\frac{A_\rho}{Z_\rho^2} = -\partial_a^2 L_\rho, \qquad \frac{B_\rho}{Z_\rho^2} = 1 + \rho\,\partial_a \Theta_\lambda L_\rho,$$

$$\frac{C_{\rho,\kappa}}{\psi_1(a) Z_\rho^2} = \frac{1}{\psi_1(a)} + \kappa\rho\,\Theta_\lambda L_\rho - \rho^2 \Theta_\lambda^2 L_\rho, \qquad \Delta_{\rho,\kappa} = A_\rho C_{\rho,\kappa} - \psi_1(a) B_\rho^2.$$

For which pairs $(\rho,\kappa)$ does $\Delta_{\rho,\kappa}$ have strictly positive Maclaurin coefficients in every positive degree for all $a > 0$; and for each $\rho > 0$, what is the sharp threshold in $\kappa$? At $\rho = 1$ the present theorem gives exactly $\kappa \ge 1$, and its proof makes essential use of the exact $\rho = 1$ Chu–Vandermonde convolution.

**Question 10.2** (Negative-order classification)**.** Under the convention of Definition 8.1, determine all real $\nu$ for which (2.14) holds for every $z > 0$ on each connected component of $\{z > 0 : I_\nu(z) \ne 0\}$. Coefficientwise positivity of the continued determinant $\Delta$ fails throughout $-3 < \nu < -2$, and pointwise failure of the Bessel inequality occurs near every negative integer $-n$, $n \ge 2$. The exceptional order $\nu = -1$ and the intervening nonintegral ranges remain open.

**Question 10.3** (Structural origin from total positivity)**.** Does there exist a canonical normalization or a bordered, possibly nonlocal, extension of the kernel $(x,s) \mapsto I_s(x)$ for which total positivity, or the positivity of an explicit confluent or bordered minor, is equivalent to the inequality $D_\nu(z) > 0$ of Theorem 2.3?

Relatedly, can the coefficientwise positivity of $AC - \psi_1(a)B^2$ be deduced formally from the scalar reciprocal-gamma log-concavity theorems of Kalmykov and Karp [8, 11]? As shown in Section 9, the Bessel–Schur determinant is not invariant under the positive row and column rescalings that preserve total-positivity signs. Any such derivation must therefore supply additional Bessel-specific input, for example through a canonical normalization, a bordered construction, or a nonlocal constraint.

## Appendix A. Confluent minors of the modified Bessel kernel

**Proposition A.1** (Confluent consequences of total positivity)**.** *With $F(s,y) = \log I_s(e^y)$, $p = F_{sy}$, and $G = -F_{ss}$ as in Section 9, the ordinary confluent limits of the total-positivity minors of $(x,s) \mapsto I_s(x)$ satisfy $p \ge 0$ and $p + sG_y \ge 0$ for $s \ge 0$, where at $s = 0$ the confluent limits are taken one-sidedly.*

*Proof.* Since $y \mapsto e^y$ is strictly increasing, replacing the $x$-variable by $x = e^y$ preserves the ordering of the nodes and hence every total-positivity sign; thus the reparametrized kernel $(y,s) \mapsto I_s(e^y)$ is totally positive. For $m \ge 1$, let $s_1 < \cdots < s_m$ and $y_1 < \cdots < y_m$. The standard multivariate divided-difference limit is

$$\begin{aligned} &\lim_{\substack{s_1,\dots,s_m \to s \\ y_1,\dots,y_m \to y}} \frac{\det\left[I_{s_i}(e^{y_j})\right]_{i,j=1}^m}{\displaystyle\prod_{1\le i<j\le m}(s_j - s_i) \prod_{1\le i<j\le m}(y_j - y_i)} \\ &= \frac{\det\left[\partial_s^{i-1}\partial_y^{j-1} I_s(e^y)\right]_{i,j=1}^m}{\displaystyle\prod_{r=0}^{m-1}(r!)^2}. \end{aligned} \tag{A.1}$$

The limits are taken through increasing nodes, so total positivity makes the left-hand side nonnegative. At the boundary $s = 0$, the $s$-nodes approach one-sidedly from $s \ge 0$; analyticity in the order identifies the resulting limits with the displayed order derivatives. At order two, (A.1) gives

$$\det\begin{pmatrix} I_s(e^y) & \partial_y I_s(e^y) \\ \partial_s I_s(e^y) & \partial_s\partial_y I_s(e^y) \end{pmatrix} = I_s(e^y)^2 p,$$

so $p \ge 0$ after division by $I_s(e^y)^2 > 0$. By (A.1), at order three the Vandermonde-normalized confluent limit is

$$\frac{1}{(0!1!2!)^2} \det\left[\partial_s^{i-1}\partial_y^{j-1} I_s(e^y)\right]_{i,j=1}^3 = \frac{1}{4}\det\left[\partial_s^{i-1}\partial_y^{j-1} I_s(e^y)\right]_{i,j=1}^3.$$

Factoring $e^F = I_s(e^y)$ from each row and performing elementary row and column eliminations gives

$$\begin{aligned} &\frac{1}{I_s(e^y)^3}\det\left[\partial_s^{i-1}\partial_y^{j-1} I_s(e^y)\right]_{i,j=1}^3 \\ &= 2F_{sy}^3 + F_{sy}F_{ssyy} - F_{ssy}F_{syy} = 2p^3 - pG_{yy} + p_yG_y. \end{aligned}$$

After additionally dividing by the positive factor $I_s(e^y)^3$, the order-three confluent limit has the same sign, so

$$2p^3 - pG_{yy} + p_yG_y \ge 0.$$

The logarithmic Bessel equation

$$F_{yy} + F_y^2 = e^{2y} + s^2,$$

differentiated once in $s$, gives $p_y + 2F_yp = 2s$; differentiated again, and using $p_s = F_{ssy} = -G_y$, it gives $G_{yy} = 2p^2 - 2F_yG_y - 2$. Substituting these into the Wronskian expression,

$$2p^3 - pG_{yy} + p_yG_y = 2(p + sG_y),$$

so the order-three confluent limit gives $p + sG_y \ge 0$. □

## Computational verification

The proofs herein do not depend on numerical computation. Verification scripts are available at `https://github.com/johnnyshields/papers`.

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## Declaration of generative AI use

During the preparation of this work, the author used OpenAI's GPT-5.6 Sol and Anthropic's Claude Opus 4.8 to assist with manuscript drafting and verification code development. The author takes full responsibility for the content of the work.

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